

Brownian Motion, Stopping Times, and Quadratic Variation

Lecture 2 Notes — Class Support and Student Summary

1 From Random Walks to Brownian Motion

1.1 Discrete intuition and scaling

Let $\varepsilon_1, \varepsilon_2, \dots$ be independent with $\mathbb{P}(\varepsilon_k = 1) = \mathbb{P}(\varepsilon_k = -1) = 1/2$, and set

$$S_n = \sum_{k=1}^n \varepsilon_k.$$

Then $\mathbb{E}[S_n] = 0$ and $\text{Var}(S_n) = n$. A typical displacement therefore has order $\sqrt{n}$, not order n .

Proposition 1.1 (Mean and variance of the random walk). For every $n \geq 0$,

$$\mathbb{E}[S_n] = 0, \quad \text{Var}(S_n) = n.$$

Proof. Linearity gives $\mathbb{E}[S_n] = \sum_k \mathbb{E}[\varepsilon_k] = 0$. Independence gives $\text{Var}(S_n) = \sum_k \text{Var}(\varepsilon_k) = n$. □

To fit n shocks into one unit of time while keeping a non-degenerate variance, define

$$W_t^{(n)} = \frac{S_{\lfloor nt \rfloor}}{\sqrt{n}},$$

with linear interpolation between grid points. For grid times $s < t$,

$$\text{Var}(W_t^{(n)} - W_s^{(n)}) = t - s.$$

Theorem 1.2 (Donsker's invariance principle, informal). On every bounded time interval, the rescaled random-walk path $W^{(n)}$ converges in distribution to a standard Brownian motion as $n \rightarrow \infty$.

This is the functional counterpart of the central limit theorem: many small, centered, independent shocks produce a Gaussian continuous-time limit.

Exercise — why the square-root normalization is unique

Suppose each of the n random-walk shocks is multiplied by $n^{-\alpha}$. Find the variance after one unit of time. For which α does it converge to a finite nonzero limit?

Short solution

The variance is $n(n^{-\alpha})^2 = n^{1-2\alpha}$. It has a finite nonzero limit only when $1 - 2\alpha = 0$, hence $\alpha = 1/2$.

2 Brownian Motion and Gaussian Structure

2.1 Definition and finite-dimensional laws

Definition 2.1 (Standard Brownian motion). A process $W = (W_t)_{t \geq 0}$ is a standard Brownian motion if:

(BM1) $W_0 = 0$ almost surely;

(BM2) increments over disjoint intervals are independent;

(BM3) $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$;

(BM4) its paths are continuous almost surely.

Continuity means no jumps. It does not imply smoothness: Brownian paths are extremely irregular at every time scale.

Definition 2.2 (Gaussian vector and Gaussian process). A vector $Z \in \mathbb{R}^m$ is Gaussian if $a^\top Z$ is Gaussian for every $a \in \mathbb{R}^m$. A process X is Gaussian if $(X_{t_1}, \dots, X_{t_m})$ is Gaussian for every finite collection of times.

Proposition 2.3 (Brownian motion is a centered Gaussian process). For every $0 < t_1 < \dots < t_m$, the vector $(W_{t_1}, \dots, W_{t_m})$ is centered Gaussian.

Proof. Let $\Delta_j = W_{t_j} - W_{t_{j-1}}$, with $t_0 = 0$. The vector $(\Delta_1, \dots, \Delta_m)$ has independent Gaussian coordinates, and

$$W_{t_k} = \sum_{j=1}^k \Delta_j.$$

The level vector is therefore a linear transformation of a Gaussian vector. □

Proposition 2.4 (Brownian covariance kernel). For $s, t \geq 0$,

$$\mathbb{E}[W_t] = 0, \quad K(s, t) := \text{Cov}(W_s, W_t) = \min(s, t).$$

The kernel K is positive semidefinite.

Proof. If $s \leq t$, write $W_t = W_s + (W_t - W_s)$. The second term is centered and independent of W_s , so $\text{Cov}(W_s, W_t) = \text{Var}(W_s) = s$. Moreover,

$$\sum_{i,j} a_i a_j K(t_i, t_j) = \text{Var}\left(\sum_i a_i W_{t_i}\right) \geq 0.$$

□

A Gaussian law is determined by its mean vector and covariance matrix. Consequently, the zero mean and the kernel $K(s, t) = \min(s, t)$ determine all finite-dimensional Brownian laws.

Example — a symbolic covariance matrix

For $0 < r < s < t$,

$$\text{Cov}((W_r, W_s, W_t)^\top) = \begin{pmatrix} r & r & r \\ r & s & s \\ r & s & t \end{pmatrix}.$$

The repeated entries represent shared history.

2.2 Transformations and path properties

Proposition 2.5 (Reusable Brownian transformations). Fix $c > 0$ and $s > 0$. The processes

$$(-W_t)_{t \geq 0}, \quad \left(\frac{W_{ct}}{\sqrt{c}}\right)_{t \geq 0}, \quad (W_{s+t} - W_s)_{t \geq 0}$$

are standard Brownian motions. Moreover,

$$(W_{s-t} - W_s)_{0 \leq t \leq s}$$

has the law of Brownian motion restricted to $[0, s]$.

Proof. Each process starts from zero and is continuous. Its increments are linear transforms of Brownian increments over disjoint intervals, hence remain independent and Gaussian. Their variances are respectively $t - u$, $(c(t - u))/c = t - u$, and $t - u$. These are the four defining properties. □

The scaling property gives the square-root-of-time rule:

$$W_{ct} \stackrel{d}{=} \sqrt{c} W_t.$$

Theorem 2.6 (Brownian roughness). Almost surely, a Brownian path is nowhere differentiable and has infinite total variation on every nontrivial interval.

The useful heuristic is

$$\frac{W_{t+h} - W_t}{h} \sim \mathcal{N}(0, 1/h).$$

The variance of the difference quotient explodes as $h \downarrow 0$, so a stable tangent is impossible. These pathwise theorems are stated without proof here.

Proposition 2.7 (Oscillation and the law of the iterated logarithm). Almost surely,

$$\limsup_{t \rightarrow \infty} W_t = +\infty, \quad \liminf_{t \rightarrow \infty} W_t = -\infty,$$

and

$$\begin{aligned} \limsup_{t \rightarrow \infty} \frac{W_t}{\sqrt{2t \log \log t}} &= 1, & \liminf_{t \rightarrow \infty} \frac{W_t}{\sqrt{2t \log \log t}} &= -1, \\ \limsup_{t \downarrow 0} \frac{W_t}{\sqrt{2t \log \log(1/t)}} &= 1, & \liminf_{t \downarrow 0} \frac{W_t}{\sqrt{2t \log \log(1/t)}} &= -1. \end{aligned}$$

Exercise — time reversal

Fix $s > 0$ and set $R_t = W_{s-t} - W_s$ for $0 \leq t \leq s$. Verify directly the distribution of $R_t - R_u$ for $u < t$, and explain why increments over disjoint intervals are independent.

Short solution

$$R_t - R_u = W_{s-t} - W_{s-u} \sim \mathcal{N}(0, t - u).$$

Disjoint intervals in reversed time correspond to disjoint intervals of the original Brownian path. Continuity is inherited, so R has Brownian law on $[0, s]$.

3 Markov Renewal and Martingales

Let $\mathcal{F}_t = \sigma(W_u : 0 \leq u \leq t)$, completed in the usual way. It represents the entire observed Brownian path up to time t .

Proposition 3.1 (Deterministic-time renewal). For fixed $t \geq 0$,

$$\widetilde{W}_u = W_{t+u} - W_t, \quad u \geq 0,$$

is a standard Brownian motion independent of $\mathcal{F}_t$.

Proof. The Brownian properties follow from stationary independent increments and continuity. Every increment of $\widetilde{W}$ occurs after time t , so it is independent of the past $\mathcal{F}_t$. $\square$

Proposition 3.2 (Markov property in conditional-expectation form). For every bounded measurable f and $h > 0$,

$$\mathbb{E}[f(W_{t+h}) \mid \mathcal{F}_t] = \mathbb{E}[f(W_{t+h}) \mid W_t] = (P_h f)(W_t),$$

where

$$(P_h f)(x) = \int_{\mathbb{R}} f(x+y) \frac{e^{-y^2/(2h)}}{\sqrt{2\pi h}} dy.$$

Proof. Write $W_{t+h} = W_t + \widetilde{W}_h$, where $\widetilde{W}_h \sim \mathcal{N}(0, h)$ is independent of $\mathcal{F}_t$. When conditioning on $\mathcal{F}_t$, W_t is known and the remaining expectation is the displayed Gaussian integral. $\square$

The operators $(P_h)_{h \geq 0}$ form a semigroup:

$$P_{h+k} = P_h P_k.$$

This follows either from Gaussian convolution or by applying the Markov property twice.

Proposition 3.3 (Brownian motion is a martingale). For $0 \leq s < t$,

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s.$$

Proof. Use $W_t = W_s + (W_t - W_s)$. The increment is centered and independent of $\mathcal{F}_s$. $\square$

Markov and martingale are different statements. Markov says that the current state summarizes the past for the entire future law. Martingale says only that the current value is the conditional mean of a future value.

Example — a conditional Gaussian moment

For $\theta \in \mathbb{R}$, renewal gives

$$\mathbb{E}[e^{\theta W_{t+h}} \mid \mathcal{F}_t] = e^{\theta W_t} \mathbb{E}[e^{\theta \tilde{W}_h}] = \exp\left(\theta W_t + \frac{1}{2}\theta^2 h\right).$$

Exercise — conditional second moment

For $0 \leq s < t$, compute $\mathbb{E}[W_t^2 \mid \mathcal{F}_s]$. Deduce a second Brownian martingale.

Short solution

Expanding $W_t = W_s + (W_t - W_s)$ gives

$$\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t - s).$$

Hence $\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s$, so $W_t^2 - t$ is a martingale.

4 Stopping Times and Strong Markov Property

4.1 No look-ahead

In discrete time, a random integer τ is a stopping time when $\{\tau \leq n\} \in \mathcal{F}_n$ for every n . A first hit can be detected as the path unfolds; a last hit before a future horizon cannot.

Definition 4.1 (Stopping time). A random time $\tau : \Omega \rightarrow [0, \infty]$ is a stopping time for $(\mathcal{F}_t)$ if

$$\{\tau \leq t\} \in \mathcal{F}_t \quad \text{for every } t \geq 0.$$

The event $\{\tau \leq t\}$ asks whether the rule has already triggered. The answer may use the observed path through time t , but not the future.

Proposition 4.2 (Canonical examples). The following are stopping times:

$$t_0, \quad \tau_a = \inf\{t \geq 0 : W_t = a\}, \quad \tau_{\ell,u} = \inf\{t \geq 0 : W_t \notin (\ell, u)\}.$$

For fixed $T > 0$, the last zero $\rho_T = \sup\{t \leq T : W_t = 0\}$ and the time of the maximum on $[0, T]$ are not stopping times.

Proof. The deterministic time is immediate. By continuity,

$$\{\tau_a \leq t\} = \left\{ \max_{0 \leq v \leq t} W_v \geq a \right\}$$

for $a > 0$, and the right-hand event is observable by time t . The same argument works for first exit. Identifying a last zero or the global maximum requires checking that no later event occurs, which uses future information. $\square$

Definition 4.3 (Information available at a stopping time). For a stopping time τ ,

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for every } t \geq 0\}.$$

An event in $\mathcal{F}_\tau$ can be decided using only information revealed by the moment the rule triggers. For continuous adapted W and almost surely finite τ , the stopped value W_τ is $\mathcal{F}_\tau$ -measurable.

Theorem 4.4 (Strong Markov property). If τ is an almost surely finite stopping time, then

$$(W_{\tau+u} - W_\tau)_{u \geq 0}$$

is a standard Brownian motion independent of $\mathcal{F}_\tau$.

Remark 4.5 (Proof idea). Round τ upward to dyadic stopping times $\tau_n = 2^{-n} \lceil 2^n \tau \rceil$. On each event $\{\tau_n = k2^{-n}\}$, apply deterministic renewal. Then let $n \rightarrow \infty$ and use path continuity. The full measure-theoretic proof is not needed in this lecture.

Exercise — first hit after a deterministic date

Fix $s > 0$ and define

$$\tau = \inf\{t \geq s : W_t = 0\}.$$

Is τ a stopping time? Contrast it with the last zero before a fixed horizon T .

Short solution

Yes. By time t , one can inspect the observed path on $[s, t]$ and decide whether a zero occurred. The last zero before T is not a stopping time, because at time $t < T$ one cannot rule out a later return to zero.

5 Reflection Principle and Hitting Probabilities

Let

$$M_t = \max_{0 \leq u \leq t} W_u, \quad \tau_a = \inf\{u \geq 0 : W_u = a\}, \quad a > 0.$$

By continuity, $\{M_t \geq a\} = \{\tau_a \leq t\}$.

On the event $\{\tau_a \leq t\}$, define the reflected path

$$W_u^* = \begin{cases} W_u, & u \leq \tau_a, \\ 2a - W_u, & u > \tau_a. \end{cases}$$

Strong Markov says that the continuation after τ_a is fresh Brownian motion; symmetry says that its negative has the same law. Therefore reflection preserves Brownian probability.

Theorem 5.1 (Reflection principle). For $a > 0$ and $t > 0$,

$$\mathbb{P}(M_t \geq a) = 2\mathbb{P}(W_t \geq a) = 2 \left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right) \right).$$

Proof. Split $\{M_t \geq a\}$ according to the terminal value:

$$\{M_t \geq a\} = \{W_t \geq a\} \cup \{M_t \geq a, W_t < a\}.$$

Reflection after τ_a is a bijection from the second event to $\{W_t > a\}$ and preserves probability. Since $\mathbb{P}(W_t = a) = 0$, the two pieces have the same probability. $\square$

Corollary 5.2 (First-passage distribution and density). For $a > 0$,

$$\mathbb{P}(\tau_a \leq t) = 2 \left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right) \right),$$

and

$$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi t^3}} \exp\left(-\frac{a^2}{2t}\right), \quad t > 0.$$

Corollary 5.3 (Maximum and scaling of the hitting time).

$$M_t \stackrel{d}{=} |W_t|, \quad \mathbb{E}[M_t] = \sqrt{\frac{2t}{\pi}}, \quad \tau_a \stackrel{d}{=} a^2 \tau_1.$$

Moreover, $\mathbb{P}(\tau_a < \infty) = 1$, while $\mathbb{E}[\tau_a] = \infty$.

The infinite mean follows from

$$\mathbb{P}(\tau_a > t) = 2\Phi(a/\sqrt{t}) - 1 \sim a\sqrt{\frac{2}{\pi t}},$$

whose integral diverges at infinity.

Example — a symbolic barrier probability

Suppose a driftless log-price satisfies $\log(S_t/S_0) = \sigma W_t$, and let $H > S_0$. Then

$$\mathbb{P}\left(\max_{0 \leq u \leq T} S_u \geq H\right) = 2 \left[1 - \Phi\left(\frac{\log(H/S_0)}{\sigma\sqrt{T}}\right)\right].$$

Reflection converts a path-dependent barrier event into a Gaussian tail.

Exercise — distribution and mean of the maximum

Derive $\mathbb{P}(M_t \leq a)$ for $a \geq 0$, and compute $\mathbb{E}[M_t]$.

Short solution

$$\mathbb{P}(M_t \leq a) = 2\Phi(a/\sqrt{t}) - 1.$$

Thus $M_t \stackrel{d}{=} |W_t|$, and the mean of a half-normal variable is $\mathbb{E}[M_t] = \sqrt{2t/\pi}$.

6 Quadratic Variation and Brownian Roughness

6.1 Definition and basic properties

For a partition $\pi = \{0 = t_0 < \dots < t_n = t\}$, let $|\pi| = \max_i(t_{i+1} - t_i)$.

Definition 6.1 (Quadratic variation). For a continuous process X , define

$$[X]_t = \lim_{|\pi| \rightarrow 0} \sum_i (X_{t_{i+1}} - X_{t_i})^2,$$

when the limit exists in probability.

Quadratic variation measures accumulated fine-scale fluctuation. It ignores the direction of increments and is different from ordinary total variation.

Proposition 6.2 (Finite-variation paths have zero quadratic variation). If A is continuously differentiable on $[0, T]$, then $[A]_T = 0$. More generally, the result holds for continuous finite-variation paths.

Proof. For $A \in C^1$, the mean value theorem gives

$$\sum_i (\Delta_i A)^2 \leq \|A'\|_\infty^2 \sum_i (\Delta_i t)^2 \leq \|A'\|_\infty^2 T |\pi| \rightarrow 0.$$

□

Theorem 6.3 (Brownian quadratic variation). For every $T \geq 0$,

$$[W]_T = T.$$

More precisely, for every partition π of $[0, T]$,

$$\mathbb{E} \left[\left(\sum_i (\Delta_i W)^2 - T \right)^2 \right] \leq 2T |\pi|.$$

Proof. The independent increments satisfy $\Delta_i W \sim \mathcal{N}(0, \Delta_i t)$. Hence

$$\mathbb{E} \left[\sum_i (\Delta_i W)^2 \right] = \sum_i \Delta_i t = T.$$

For $G \sim \mathcal{N}(0, v)$, $\text{Var}(G^2) = 2v^2$. Therefore

$$\text{Var} \left(\sum_i (\Delta_i W)^2 \right) = 2 \sum_i (\Delta_i t)^2 \leq 2|\pi| \sum_i \Delta_i t = 2T|\pi|.$$

The realized sum converges to T in L^2 , hence in probability. $\square$

Proposition 6.4 (Drift disappears from quadratic variation). If A is continuously differentiable and $X = A + \sigma W$, then

$$[X]_t = \sigma^2 t.$$

In particular, $[aX]_t = a^2[X]_t$ and $[X + A]_t = [X]_t$.

Proof. Expand the squared increments. The sum of $(\Delta A)^2$ vanishes. For the cross-term, Cauchy–Schwarz gives

$$\left| \sum_i \Delta_i A \Delta_i W \right| \leq \left(\sum_i (\Delta_i A)^2 \right)^{1/2} \left(\sum_i (\Delta_i W)^2 \right)^{1/2},$$

which tends to zero in probability. The Brownian term tends to $\sigma^2 t$. $\square$

Process	Quadratic variation	Fine-scale component
smooth or finite-variation A	$[A]_t = 0$	none
Brownian motion W	$[W]_t = t$	one variance unit per unit time
$A_t + \sigma W_t$	$[X]_t = \sigma^2 t$	volatility, not drift

6.2 The bracket and the new calculus

Proposition 6.5 (Quadratic-variation martingale). For a continuous square-integrable martingale X , there is a unique increasing predictable process $\langle X \rangle$, starting from zero, such that

$$X_t^2 - \langle X \rangle_t \text{ is a martingale.}$$

For continuous martingales, $\langle X \rangle = [X]$. In particular, $\langle W \rangle_t = t$.

For Brownian motion this can be checked without general theory:

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 + (t - s) - t = W_s^2 - s.$$

The Itô multiplication rules summarize the scale of the increments:

$$(dW_t)^2 = dt, \quad dW_t dt = 0, \quad (dt)^2 = 0.$$

These are shorthand for limiting statements about quadratic variation, not ordinary algebraic identities.

Exercise — realized variance with drift

Let $X_t = A_t + \sigma W_t$, where $A \in C^1$, and observe X on a partition of $[0, T]$. Show that

$$\hat{\sigma}_\pi^2 = \frac{1}{T} \sum_i (X_{t_{i+1}} - X_{t_i})^2$$

converges in probability to σ^2 as $|\pi| \rightarrow 0$.

Short solution

The numerator is the realized quadratic variation of X , which converges to $[X]_T = \sigma^2 T$. Division by T gives the result.

Numerical interpretation

On observed data, $\sum_i (\Delta_i X)^2$ is the realized variance. Refining the grid removes smooth drift and reveals accumulated volatility. In actual market data, extremely fine grids may be contaminated by bid-ask bounce and other microstructure noise, so “finer” is not automatically “better.”

7 Geometric Brownian Motion

An additive model $S_t = S_0 + \mu t + \sigma W_t$ can become negative and has constant absolute volatility. A multiplicative model instead makes log-returns Gaussian.

Definition 7.1 (Geometric Brownian motion). For $S_0 > 0$ and $\sigma > 0$,

$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right).$$

Then $S_t > 0$ for every t .

Proposition 7.2 (Laws and moments of GBM). For $0 \leq s < t$,

$$\log(S_t/S_s) \mid \mathcal{F}_s \sim \mathcal{N}\left((\mu - \frac{1}{2}\sigma^2)(t-s), \sigma^2(t-s)\right),$$

independently of $\mathcal{F}_s$. Moreover,

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \text{Var}(S_t) = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1),$$

and

$$\text{Median}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}.$$

Proof. The log-return is a deterministic drift plus $\sigma(W_t - W_s)$, giving the conditional normal law. For $G \sim \mathcal{N}(0, t)$, $\mathbb{E}[e^{\lambda G}] = e^{\lambda^2 t/2}$. Applying this with $\lambda = \sigma$ and 2σ gives the first two moments and hence the variance. The median is obtained by setting the centered Gaussian term to its median zero. $\square$

For $K > 0$,

$$\mathbb{P}(S_t > K) = 1 - \Phi\left(\frac{\log(K/S_0) - (\mu - \sigma^2/2)t}{\sigma\sqrt{t}}\right).$$

Proposition 7.3 (Quadratic variation of the log-price).

$$[\log S]_t = \sigma^2 t.$$

Proof. The log-price is a continuously differentiable drift plus σW_t . The drift has zero quadratic variation and zero limiting cross-term with the Brownian part. $\square$

GBM is tractable and preserves positivity, but it omits jumps, heavy tails, changing volatility, and volatility smiles. Brownian motion is a foundational building block rather than a complete market description.

What to retain

Brownian motion is a centered Gaussian process with covariance $\min(s, t)$, fresh increments, and continuous rough paths. Markov renewal describes what happens after a current or stopping time; reflection converts barrier events into Gaussian tails; quadratic variation records accumulated volatility and creates the correction terms of stochastic calculus.